



# Formal Language & Finite Automata

## Recap

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## Context-free languages - examples

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## CFG's and PDA's

Consider the following context-free grammar  $G = (V, \Sigma, R, S)$ , where  $V = \{S, A, B\}$ ,  $\Sigma = \{0, 1\}$  and the production rules are

$$S \rightarrow BSA$$

$$A \rightarrow 1$$

$$B \rightarrow 01$$

$$S \rightarrow 011$$

1. Describe in natural language the strings generated by the grammar  $G$ .
2. Produce a parse tree from the string 010101111.
3. Construct a pushdown automaton that recognises  $L(G)$ .

## Closure properties

Let  $A_1$ ,  $A_2$ , and  $A_3$  be languages defined over the alphabet  $\Sigma = \{a, b\}$ , where

1.  $A_1$  consists of all possible strings over  $\Sigma$  except the strings  $w_1, w_2, \dots, w_{25} \in \Sigma^*$ .
2.  $A_2$  is generated by the context-free grammar  $J$ .
3.  $A_3$  is recognised by the push down automaton  $M$ .

Prove that  $A_1 \cap (A_3 \cup A_2)$  is a context-free language.

## Context-free vs. regular languages

Consider the following language,

$$L = \{xy \mid x \in \{a, b\}^* a \{a, b\}^*, y \in \{a, b\}^* b \{a, b\}^* \text{ and } |x| = |y|\}.$$

Prove that  $L$  is a context-free language and that  $L$  is not a regular language.

## Reminder pumping lemma for regular languages

### Theorem (Pumping lemma for regular languages)

*If  $A$  is a regular language, then there is a number  $p$  where if  $s$  is any string in  $A$  of length at least  $p$ , then  $s$  may be divided into three parts,  $s = xyz$ , satisfying the following conditions:*

1. *for each  $i \geq 0$ ,  $xy^iz \in A$ ,*
2.  *$|y| > 0$ , and*
3.  *$|xy| \leq p$ .*